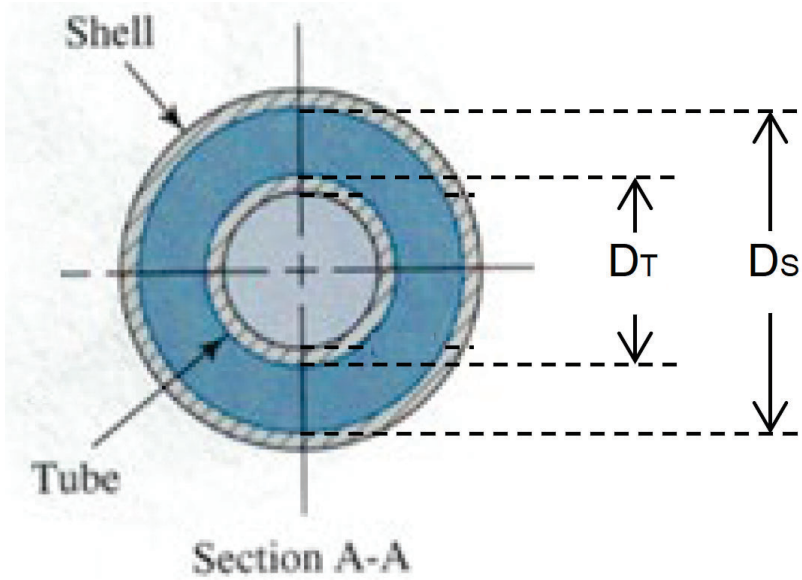
Tutorial Problem #2 .

For flow through the annular space between the shell and tube sections shown in the accompanying figure, a colleague tells you that the hydraulic diameter is simply the difference between the internal diameter of the shell section and the outside diameter of the tube section.

Prove that your colleague is correct.



$$D_h = \frac{4A}{P}$$

$$D_h = \frac{4\left(\frac{\pi}{4}{D_S}^2 - \frac{\pi}{4}{D_T}^2\right)}{\pi D_S + \pi D_T}$$

$$D_h = \frac{\left({D_S}^2 - {D_T}^2\right)}{D_S + D_T}$$

$$D_h = \frac{\left(D_S - D_T\right)\left(D_S + D_T\right)}{D_S + D_T}$$

$$D_h = D_S - D_T$$